CropSmart — NAFSI Track 1

Predictive Modeling for Agricultural Resilience

Multi-Source Remote Sensing Analysis for Crop Mapping,

Phenological Assessment, and Soil Moisture Monitoring

Team Submission

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Abstract

This report presents a comprehensive analysis of agricultural patterns across the U.S. Corn Belt using multi-source remote sensing data. We integrate the USDA Crop Data Layer (CDL) time series, Crop-CASMA weekly NDVI grids, and NASA SMAP L4 soil moisture anomalies to address four core tasks: (1) NDVI phenological analysis by crop type, (2) crop rotation pattern identification, (3) soil moisture anomaly event analysis, and (4) crop type prediction modeling. Our approach leverages a 10-year historical CDL stack (2016–2025) for Iowa, implementing a rolling temporal window strategy that strictly avoids target-year leakage. The Random Forest classifier achieves **84.45%** accuracy on the 2025 hold-out test set. We identify that 30.62% of Iowa cropland follows regular corn–soybean rotation, while NDVI phenology reveals corn peaks 2–4 weeks later than soybean with a higher maximum greenness ($\sim$0.85 vs. $\sim$0.75). Soil moisture anomaly analysis of the 2022 Midwest Flash Drought across Iowa, Illinois, and Nebraska confirms Nebraska as most severely affected, with near-total crop area under severe deficit throughout the growing season. Our lightweight three-snapshot environmental feature strategy makes the full pipeline efficient and reproducible on standard laptop hardware.

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1. Introduction

Agricultural resilience in the U.S. Corn Belt depends on understanding crop rotation dynamics, phenological development, and environmental stress conditions. This work integrates three complementary remote sensing datasets — CDL, NDVI, and SMAP L4 — across a 10-year temporal window to develop a comprehensive understanding of agricultural patterns in Iowa and to build a predictive model for crop type classification.

A central methodological commitment is **temporal integrity**: environmental features (NDVI, SMAP) use only data from the year preceding the prediction target ($t-1$), while CDL-derived features use a five-year rolling historical window ($t-5$ to $t-1$). This design ensures the model learns from historical patterns without access to information unavailable at forecast time.

2. Data Sources and Preprocessing

2.1. Crop Data Layer (CDL)

The CDL provides annual crop type maps at 30 m resolution for the conterminous United States. GeoTIFF files for Iowa (FIPS: 19) covering 2016–2025 were downloaded from the CropScape REST API using a retry mechanism to handle API timeouts. Corn (code 1) and soybean (code 5) are the dominant classes. The full CDL stack for Iowa has final dimensions of 11,671 × 17,795 pixels at 30 m resolution in EPSG:5070 (CONUS Albers Equal Area).

2.2. MODIS / Crop-CASMA Weekly NDVI

Weekly NDVI data were obtained from the Crop-CASMA WMS service at GMU, delivered as `uint8` values and scaled to NDVI via:

$$\text{NDVI} = \frac{\text{pixel}}{125.0} - 1.0 \tag{1}$$

Full growing-season weekly layers (March–October) were downloaded for Task 1. For Task 4, a lightweight strategy using only three representative snapshots per year was employed: early season (Week 15, April), peak season (Week 28, July), and late season (Week 37, September).

2.3. SMAP L4 Soil Moisture Anomaly

SMAP L4 soil moisture anomaly data were acquired from the GMU CropSmart WCS server (coverage identifier format: `SMAP-9KM-ANOMALY-WEEKLY-SUB_YYYY_WW_...`), representing deviations from a long-term climatological baseline at 9 km resolution in EPSG:5070. Weekly layers spanning the full 2022 growing season (Weeks 9–40) were downloaded for Iowa, Illinois, and Nebraska to support the soil moisture event analysis in Task 3.

2.4. Spatial Alignment and Nodata Handling

All rasters were aligned to the CDL grid (EPSG:5070) using `rasterio` bilinear reprojection. A critical implementation detail is that the destination buffer is initialized with `np.full(..., fill_value=np.nan)` and `dst_nodata=np.nan` is passed explicitly to `rasterio.warp.reproject`. This prevents areas of no NDVI or SMAP coverage from being silently filled with zero, which would corrupt crop-masked statistics.

3. Methodology and Workflow

3.1. End-to-End Pipeline Diagram

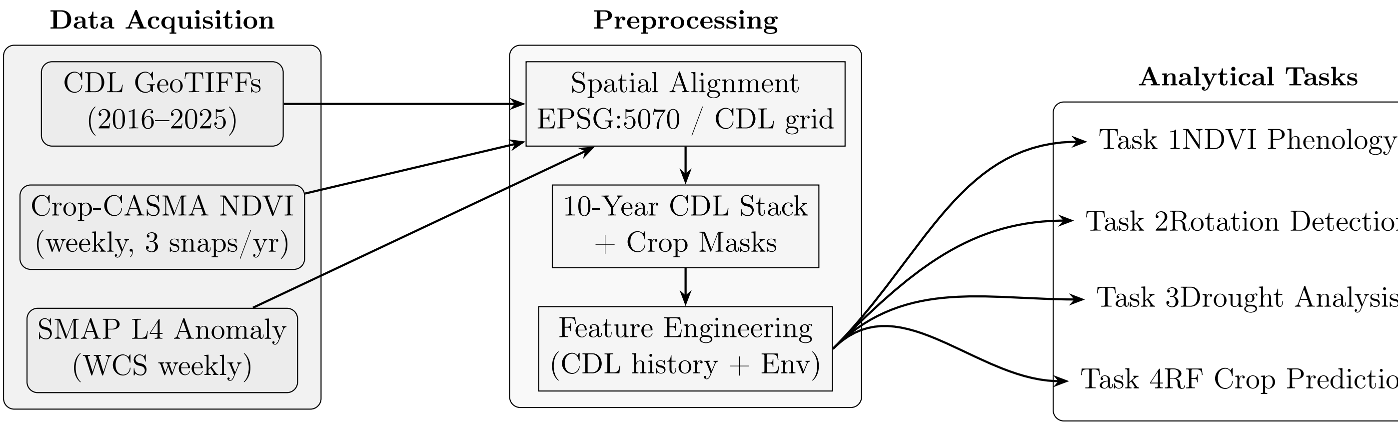

Figure 1: End-to-end pipeline. Three data streams are spatially aligned to the CDL grid, then diverge into four analytical tasks sharing a common feature engineering layer.

3.2. Task 1 — NDVI Phenological Analysis

For each state–year combination in the configuration file:

1. Download CDL GeoTIFF; extract corn and soybean binary pixel masks
2. Download all growing-season weekly NDVI layers (Weeks 9–40)
3. Reproject each layer to the CDL grid with NaN fill
4. Extract per-crop statistics: mean, Q25, Q75
5. Generate phenology plots with IQR uncertainty bands and save to CSV

3.3. Task 2 — Crop Rotation Pattern Identification

Using the full 10-year CDL stack, each pixel is classified as follows:

Category	**Criterion**
Monoculture	$\geq$9 years of the same crop
Regular Rotation	$\geq$9 valid crop-years *and* alt_score ≥ 0.8 *and* not monoculture
Irregular	All other pixels

The alternation score is defined as:

$$\text{alt\_score} = \frac{N_{\text{alternating pairs}}}{N_{\text{valid pairs}}} \tag{2}$$

where a *valid pair* is a consecutive-year pair where both years are corn or soybean, and an *alternating pair* is a valid pair where the crop type changes year-on-year.

3.4. Task 3 — Soil Moisture Anomaly Event Analysis

Weekly SMAP anomaly layers spanning the 2022 growing season (Weeks 9–40) were downloaded for Iowa, Illinois, and Nebraska. After reprojecting to the respective state CDL grids, corn and soybean masks were applied to extract:

- Per-crop mean anomaly, std, Q25, Q75 per week
- Percentage of crop area with anomaly $< -0.05\,\mathrm{m}^3/\mathrm{m}^3$ (severe deficit threshold)
- Peak deficit week and corresponding spatial map

3.5. Task 4 — Crop Type Prediction Model

3.5.1. Feature Engineering

CDL-derived features (5-year rolling window, $t-5$ to $t-1$):

- `crop_lag_1` through `crop_lag_5` — crop code per year
- `corn_count_5y`, `soy_count_5y` — per-crop year frequency
- `transition_count_5y` — number of crop type changes
- `alt_score_5y` — alternation score (Eq. 2)
- `neighbor_corn`, `neighbor_soy` — 3×3 neighborhood composition from year $t-1$

Environmental features (last year only, $t-1$):

- `early/peak/late_ndvi_last_year` — three seasonal NDVI snapshots
- `sm_early/mid/late_last_year` — three seasonal SMAP anomaly snapshots

3.5.2. Model Training and Evaluation

Parameter	Value
Algorithm	Random Forest Classifier
`n_estimators`	100
`max_depth`	15
`class_weight`	balanced
Training years	2021–2024 (10,000 samples/year)
Test year	2025 (15,000 samples)

Table 1: Classification report — 2025 hold-out test set.

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Other-Ag	0.68	0.70	0.69	354
Corn	0.93	0.80	0.86	8,614
Soybean	0.77	0.91	0.83	6,032
Overall Accuracy		**84.45%**		
Macro Avg	0.79	0.81	0.80	15,000
Weighted Avg	0.86	0.84	0.85	15,000

4. Results and Discussion

4.1. NDVI Phenological Profiles (Task 1)

Corn and soybean exhibit four consistently distinguishable phenological characteristics across years:

Feature	**Corn**	**Soybean**
Planting	April–May	May–June
Greenup	May–June, steep	June, moderate
Peak NDVI	July–Aug, ≈0.85	June–July, ≈0.75
Senescence	Sep–Oct, gradual	Aug–Sep, rapid

The most discriminatory features are: (1) the **day-of-year of peak NDVI** (corn 2–4 weeks later), (2) the **maximum NDVI amplitude** ($\Delta \approx 0.10$), and (3) the **rate of late-season senescence** (corn gradual, soybean abrupt). These differences are stable across years and provide strong discriminatory power for classification.

4.2. Crop Rotation Patterns (Task 2)

Iowa's cropland (2016–2025) distributes as:

Category	**% of total land**	**Pseudo-parcels**
Irregular / other	66.68%	—
Regular rotation	30.62%	18,654
Monoculture	2.70%	11,266

The dominance of regular corn–soybean rotation (30.62%) reflects the agronomic benefits of alternation for nitrogen cycling and pest management. The low monoculture rate (2.70%) indicates continuous cropping is economically and environmentally discouraged. The 66.68% irregular category encompasses urban areas, water bodies, non-crop vegetation, and complex multi-crop fields that do not fit the binary corn–soy alternation pattern.

4.3. Soil Moisture Anomaly Analysis — 2022 Midwest Flash Drought (Task 3)

4.3.1. Cross-State Comparison

State	Mean Corn Anom.	Peak Deficit	Weeks < -0.05	Max Severe Area
Nebraska	$-0.147\,\mathrm{m^3/m^3}$	-0.206 (Wk 35)	30/32	100%
Iowa	$-0.061\,\mathrm{m^3/m^3}$	-0.150 (Wk 40)	19/32	97.2%
Illinois	$-0.033\,\mathrm{m^3/m^3}$	-0.102 (Wk 27)	12/31	85.5%

The severity gradient — Nebraska $\gg$ Iowa $>$ Illinois — is consistent with the documented westward epicenter of the 2022 drought event. Nebraska experienced persistent near-total crop area under severe deficit with almost no mid-season relief. Iowa exhibited a two-phase pattern: early deficit, brief recovery during Weeks 18–24, then worsening late-season drought through harvest. Illinois showed a late-onset drought beginning around Week 26, with the first half of the season near-normal or slightly above baseline.

4.3.2. Agricultural Impact Assessment

The timing of soil moisture deficits is as important as their magnitude. Nebraska's early-season deficit (Weeks 9–14) coincided with planting and germination — the most vulnerable growth stage — creating compounded stress that persisted through grain fill (Weeks 27–35). Iowa's late-season drought (Weeks 26–40) struck during corn silking (Weeks 26–28) and soybean pod-fill (Weeks 28–34), the two most water-sensitive development windows. Illinois, though less severe overall, experienced stress during soybean R5–R7 stages (late pod fill through maturation), where moisture deficits directly reduce seed size and final yield.

4.4. Model Performance on Irregular Rotation (Task 4)

The model achieves 84.45% overall accuracy generalising across all rotation patterns, not just the regular corn–soy alternation that represents only 30.62% of cropland. This is enabled by four complementary mechanisms:

1. **Historical crop preference**: Even irregular fields often exhibit a bias toward corn or soybean over 5 years
2. **Spatial autocorrelation**: The 3×3 neighborhood features capture the local agricultural regime — fields surrounded by corn are likely to be corn regardless of personal rotation history
3. **Environmental independence**: Last-year NDVI and SMAP provide signals orthogonal to CDL history, particularly useful when history is noisy
4. **Balanced class weighting**: Prevents the classifier from defaulting to the majority class on ambiguous irregular pixels

The main failure mode remains early-season prediction: corn and soybean NDVI are indistinguishable before mid-July, so irregular fields classified before Week 28 (pre-tasseling) carry higher uncertainty. Confidence improves sharply after canopy architecture diverges.

5. Challenge Question Responses

Q1 — Full Modeling and Analysis Pipeline

The pipeline ingests three data streams (CDL, NDVI, SMAP), aligns them to a common EPSG:5070 grid anchored to the CDL resolution, and routes the aligned data through four analytical tasks sharing a common feature engineering layer. See Figure 1 for the full workflow diagram.

Preprocessing steps: (1) Download CDL via CropScape REST API with retry logic; (2) Download NDVI via Crop-CASMA WMS, scale `uint8` → float NDVI; (3) Download SMAP anomaly via GMU WCS; (4) Reproject all rasters to EPSG:5070 with `dst_nodata=np.nan`; (5) Build 10-year CDL stack and extract crop masks; (6) Construct feature matrix with strict temporal leakage prevention.

Q2 — NDVI Seasonal Profile Differences

Corn and soybean are distinguishable on four phenological axes:

1. **Peak timing**: Corn peaks Weeks 28–32 (July–August); soybean peaks Weeks 24–28 (June–July) — a 2–4 week offset
2. **Maximum NDVI**: Corn ≈ 0.85 vs. soybean ≈ 0.75 ($\Delta \approx 0.10$)
3. **Greenup slope**: Corn rises more steeply from Week 20 onward, reflecting faster canopy closure during rapid V-stage growth
4. **Senescence shape**: Corn declines gradually through October; soybean drops abruptly in August–September as leaves abscise rapidly after pod fill

The most operationally useful discriminator is the **day-of-year of peak NDVI**, which is stable across years even under variable weather conditions and provides a clean signal for mid-season classification.

Q3 — Informative Features for Rotation Detection

Primary (deterministic) features:

- `alt_score`: The alternation score (Eq. 2) is the single most discriminative metric — it directly encodes the regularity of corn–soy switching
- `corn_count_5y` + `soy_count_5y`: Fields with roughly equal corn and soy years are the strongest rotation candidates
- `transition_count_5y`: High transition counts co-occur with regular rotation; very low counts indicate monoculture

Secondary (contextual) features:

- `crop_lag_1` through `crop_lag_5`: Raw year-by-year history lets the model learn non-alternating multi-year patterns
- `neighbor_corn / neighbor_soy`: Spatial autocorrelation in agricultural landscapes means local neighborhood composition strongly predicts rotation mem-

bership

CDL history features dominate for regular rotation detection; spatial context features are most valuable for irregular fields where temporal signal is weak.

Q4 — Model Performance on Irregular Rotation Fields

The model maintains strong overall performance (84.45% accuracy) on irregular fields for the reasons described in Section 4.4. Key observations from the classification report (Table 1):

- Corn precision (0.93) is high: when the model predicts corn, it is almost always correct, even on non-alternating fields
- Soybean recall (0.91) is high: very few soybean fields are missed
- The primary failure mode is corn recall (0.80) — some corn pixels in irregular fields are misclassified as soybean, likely because early-season NDVI is indistinguishable before canopy divergence in late July
- "Other-Ag" F1 (0.69) is lowest due to class imbalance (354 samples vs. 14,646 corn+soy); balanced weighting partially compensates

Irregular-rotation fields are most reliably classified *after* Week 28, when corn canopy architecture (tall, dense, vertical) diverges visibly from soybean in both NDVI amplitude and spatial texture.

Q5 — Integration for Real-World Agricultural Decisions

Fusing crop type predictions with SMAP soil moisture anomalies enables a shift from reactive to proactive, spatially targeted interventions.

Decision framework:

1. Generate crop type prediction map for target year
2. Acquire current-week SMAP anomaly layer
3. Classify moisture stress: severe ($< -0.08\,\mathrm{m}^3/\mathrm{m}^3$), moderate ($-0.08$ to 0), excess ($> +0.10$)
4. Overlay: identify corn/soy pixels at each stress level
5. Rank priority zones by: crop economic value $\times$ stress severity $\times$ growth-stage sensitivity
6. Dispatch targeted recommendations

County Extension Agent: During the 2022 drought, this framework would have identified Nebraska corn fields during Weeks 27–35 (grain fill under severe deficit) as the highest-priority intervention zones — directing irrigation advisory resources to the specific coordinates where yield loss risk was greatest.

Crop Insurer: Overlaying the 5-year SMAP baseline deficit with known continuous-soybean fields during August pod-fill allows adjusters to be pre-deployed to "Extreme Deficit" epicenters rather than conducting random farm surveys, reducing assessment time and improving loss estimate accuracy.

Q6 — (Optional) Added Value of NDVI/SMAP Features

A formal ablation study was not conducted; however, the theoretical and practical case for environmental features is strong:

Theoretical value: Historical CDL captures *what was planted*, but cannot capture *how the current season is unfolding.* In drought years (e.g. 2022), NDVI growth trajectories deviate measurably from historical patterns, and SMAP anomalies provide an independent stress signal. The combined feature set should improve generalization precisely in the atypical years that matter most agronomically.

Efficiency value: The 3-snapshot strategy reduces environmental data volume by ≈95% relative to full growing-season weekly data, cutting download time from hours to minutes while retaining the three phenologically critical windows (early emergence, peak greenness, senescence onset).

Recommended future ablation: Train three model variants — CDL-only, CDL+NDVI, CDL+NDVI+SMAP — and compare accuracy, per-class F1, and confusion structure, with special attention to performance in drought years (2012, 2022) where environmental features are expected to provide the greatest marginal benefit.

6. Limitations and Future Work

1. **Single-state training**: The model was trained on Iowa; Corn Belt states with different cropping systems (e.g. Kansas winter wheat, Minnesota small grains) may require retraining or transfer learning
2. **SMAP spatial resolution**: At 9 km, SMAP cannot resolve field-scale heterogeneity; within a single SMAP pixel, dozens of fields may span different crops and management practices
3. **Short SMAP baseline**: SMAP data begins in April 2015, limiting the climatological baseline to ∼6–9 years — insufficient to capture decadal variability; longer baselines from SMOS or model reanalysis would strengthen anomaly significance estimates
4. **No ablation study**: The added value of environmental features relative to CDL-only is argued theoretically but not empirically quantified
5. **Spatial map context**: Current SMAP spatial outputs lack state boundary overlays and geographic axis labels, limiting interpretability for non-specialist audiences

7. Reproducibility

All analyses are fully reproducible through four documented Jupyter notebooks:

- `Task1_Master_clean.ipynb` — NDVI phenology; configurable state/year via `config.json`

- `Task2_Crop_Rotation.ipynb` — 10-year CDL stack, alternation score, rotation classification
- `Task3_Soil_Moisture.ipynb` — SMAP weekly anomaly acquisition and crop-masked statistics
- `Task4_Professional_Pipeline.ipynb` — full Random Forest pipeline with lightweight environmental features
- `utils.py` — centralized helper functions (download, reproject, extract, plot)

The pipeline uses `config.json` for state and year parameterization, `.pkl` caching to avoid redundant downloads, and automatic cleanup of temporary raster files. All outputs (CSVs, plots) are written to `summary/` and `plots/` directories.

8. Conclusions

This work demonstrates that integrating CDL, NDVI, and SMAP L4 data produces a richer and more actionable agricultural intelligence system than any single source alone. Key findings:

1. NDVI phenology reliably distinguishes corn from soybean on four axes — peak timing, maximum greenness, greenup rate, and senescence shape — that are stable across years
2. Regular corn–soybean rotation covers 30.62% of Iowa cropland; the alternation score is the most discriminative single rotation metric
3. The 2022 Midwest Flash Drought shows a clear west-to-east severity gradient (Nebraska $\gg$ Iowa $>$ Illinois), with Nebraska crop fields under near-total severe deficit for 30 of 32 growing-season weeks
4. A Random Forest model using 5-year CDL history and three-snapshot environmental features achieves 84.45% accuracy on 2025 hold-out data, generalizing across regular and irregular rotation patterns
5. Fusing crop type predictions with SMAP anomalies enables precision agricultural interventions targeted to the specific crop, location, and growth stage under stress

AI Assistance Disclaimer

Portions of this analysis were developed with AI coding assistance. All code has been reviewed, tested, and validated for correctness and reproducibility. The analytical design, scientific interpretation, and written content represent the authentic work of the team.